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Fall 2026 **Course:** COMPSCI 566

Monocular Depth Estimation: Benchmarking and a Defocus-Informed Refinement

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Progress Summary

The pipeline runs end to end. Phase 1 (the learned and classical baselines) is done: Depth Anything V2 (DA-V2) is benchmarked on all 654 NYU-Depth v2 test images at three model scales (ViT-S/B/L) plus the metric indoor checkpoint, and the classical depth-from-focus (DFF) baseline (modified Laplacian focus measure, parabolic sub-frame interpolation, confidence map, all-focus guided smoothing) runs on the rendered focal stacks and all 120 DDFF 12-Scene test stacks (Hazirbas et al.). Phase 2 (the focus-informed metric refinement) is done on synthetic data: the robust scale fit and regularized local correction run on six scenes split into tuning and held-out sets, and the combined system runs qualitatively on real DDFF stacks. Preliminary ablations on the tuning scenes cover confidence weighting, the regularization and correction weights, and focal-stack size (2/3/5/10 frames).

Current Results

NYU-Depth v2 test split (654 images, Eigen crop), DA-V2 across model scales and alignment choices (Table 1):

Checkpoint	Alignment	AbsRel	RMSE (m)	δ_1
ViT-S	median scale only	0.494	2.349	0.393
ViT-S	least squares	0.062	0.297	0.956
ViT-S	robust IRLS	0.058	0.292	0.958
ViT-B	least squares	0.057	0.283	0.959
ViT-L	least squares	**0.055**	**0.277**	**0.961**
Metric-Indoor-S	none (raw metric)	0.218	0.650	0.670
Metric-Indoor-S	least squares	0.077	0.353	0.936

The affine fit matters: a median scale alone collapses accuracy (δ_1 0.39), so we model $q = s\hat{d} + t$ instead of a scale-only alignment. Robust IRLS (iteratively reweighted least squares) buys a small, consistent gain over plain least squares. Bigger backbones help less than expected (S $\rightarrow$ L gains ≈ 0.005 δ_1 at 4$\times$ the runtime on MPS), so ViT-S stays the project baseline. The metric indoor checkpoint has a real scale prior but is biased on NYU (0.218 unaligned), and even aligned, it trails the relative models.

Synthetic focal stacks (10 frames, focus 0.9–7 m, thin-lens blur model). Regularization weights were tuned on two scenes (gallery, stairs); Table 2 reports the held-out mean over the other four (corridor, objects, slanted, shelves), metric depth scored without alignment:

Method	AbsRel	RMSE (m)	δ_1
DFF raw	0.233	1.552	0.732
DFF + all-focus smoothing	0.221	1.510	0.702
DA-V2 aligned (oracle)	0.271	1.524	0.548
DA-V2 + scale fit from focus	0.308	1.614	0.459
Refined (ours)	0.259	1.483	0.575
Combined (ours)	**0.206**	**1.284**	0.704

The focus baseline is accurate where texture exists (δ_1 0.89–1.00 on textured regions) and degrades on flat surfaces, which is exactly where the confidence map reads low ($\bar{w} \approx 0.06$–0.26 textureless, 0.26–0.66 textured). All-focus guided smoothing recovers part of that loss (textureless AbsRel $0.77 \rightarrow 0.61$ on held-out scenes) but can smear confident edges into large flat regions, a known limitation of normalized convolution. On the calibrated held-out scenes (corridor, objects, shelves) the refinement has the lowest AbsRel of any non-oracle method on every split: all pixels 0.246 vs. raw 0.277, focus 0.266, scale fit 0.312; textured 0.211 vs. 0.224; textureless 0.281 vs. 0.307. Raw DFF keeps a higher δ_1 where texture exists (0.79 vs. 0.70) because its per-pixel maxima stay sharp there. The one AbsRel it loses is low-texture against oracle-aligned DA-V2 (0.281 vs. 0.265), and that fit uses ground truth. The *combined* system falls back to the focus estimate on rejected calibrations; per scene it matches the better component, and on the held-out mean it beats both (0.206 vs. 0.259 refined, 0.221 focus). DA-V2 underperforms the focus baseline on our scenes even with oracle alignment, a plausible outcome given how far the renderer sits from its training distribution, so we read this benchmark as a stress test rather than a real-data statement.

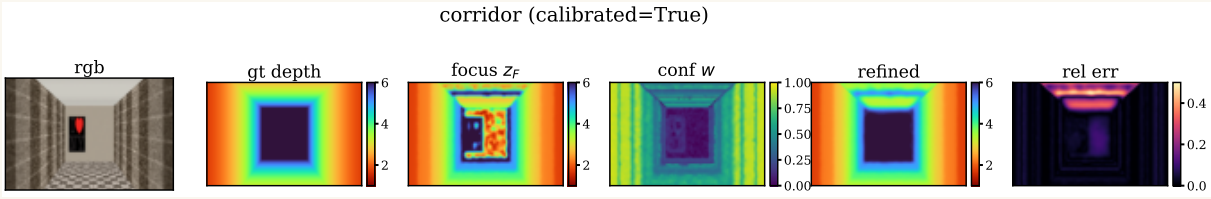


Fig. 1. Corridor, a held-out calibrated scene: input RGB, ground-truth depth, the focus estimate z_F, confidence w, our refined metric depth, and its relative error.

Difficulties

- **DDFF labels.** The public test archive ships 120 focal stacks but no ground truth (withheld for the benchmark server), and per-frame focus distances are not recorded. We reconstructed focus distances from the DDFF paper's refocus geometry (disparities linear in $[0.02, 0.28]$ px, $z \approx 0.14/d$), verified the frame-to-depth ordering per stack by Spearman correlation against the learned prior (about a third of the stacks we checked prefer a reversed ordering), and limited DDFF to qualitative evaluation. Confidences on the real Lytro stacks run low ($\bar{w} \approx 0.02$–0.05); a per-scene confidence normalization in the fit gets calibration accepted on most stacks, but none of it can be scored without labels.
- **NYU test split sourcing.** Both canonical sources were unreachable (the DenseDepth mirror returns 403 and the NYU host refuses connections); we used a mirror of the same 654-image Eigen test split and verified image count, units, and depth range.
- **Defocus veiling.** In the layered renderer, blurred near-field texture bleeds over flat surfaces

and lifts the focus measure there (the corridor's flat ceiling reads well above the textureless range). That is a genuine failure mode of the measurement, not a renderer artifact, and it caps how much we can trust w at occlusion boundaries.

- **Rejected fits.** Where the learned prediction anticorrelates with the focus estimate, the scale fit degenerates ($s \leq 0$). On those scenes the refined output falls back to a constant plane at the confidence-weighted focus mean, marked uncalibrated, rather than emit bad metric depth.

Changes from Proposal

DDFF 12-Scene moved from a quantitative to a qualitative dataset: the released test stacks carry no labels, as the proposal's planned verification found. Rendered scenes now carry the quantitative refinement evaluation, split into tuning and held-out sets so the reported numbers are not tuned on the held-out scenes. The proposal scheduled scale fitting and refinement for Oct 30–Nov 13; that stage already runs on the synthetic scenes, roughly three weeks ahead. Three additions beyond the proposed method: an all-focus guided smoothing stage on the DFF baseline (normalized convolution in inverse depth, weighted by confidence); a combined output that falls back to the focus estimate on rejected calibrations (best held-out mean, Table 2); and a degenerate-fit guard replacing silent bad calibration.

Next Steps

- Publish per-checkpoint runtimes on the webpage next to the NYU benchmark (already logged in the CSVs; the S/B/L and metric comparison is already up). Held-out evaluation and webpage work run Nov 14–24.
- Extend the DDFF qualitative evaluation to more stacks and, time permitting, phone-captured focal stacks as a second real source.
- Build the presentation deck (Dec 1–10) and keep the webpage figures in step with the final numbers.

References

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N. Silberman et al. Indoor Segmentation and Support Inference from RGBD Images (NYU-Depth v2). *ECCV*, 2012.

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Code and figures: https://github.com/DimitriChrysafis/cs566-project